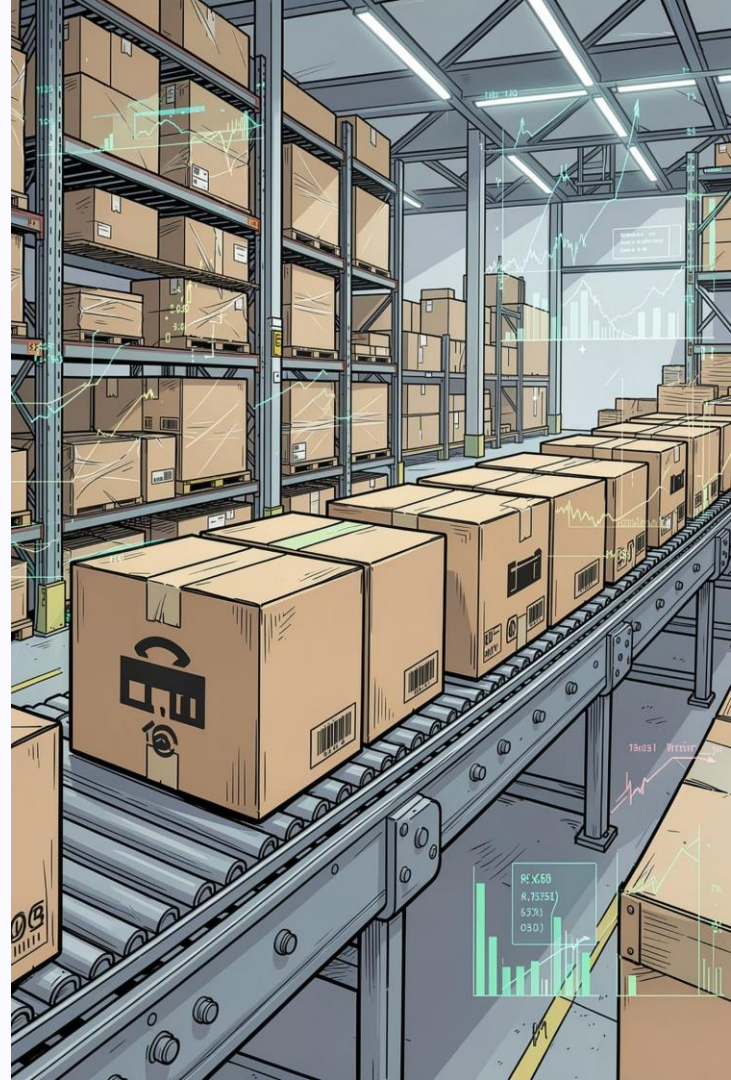


E-Commerce Funnel Analysis

End-to-end data analyst project — tracking every order from placement to delivery

OLIST DATASET · KAGGLE

SQL · PYTHON · POWER BI



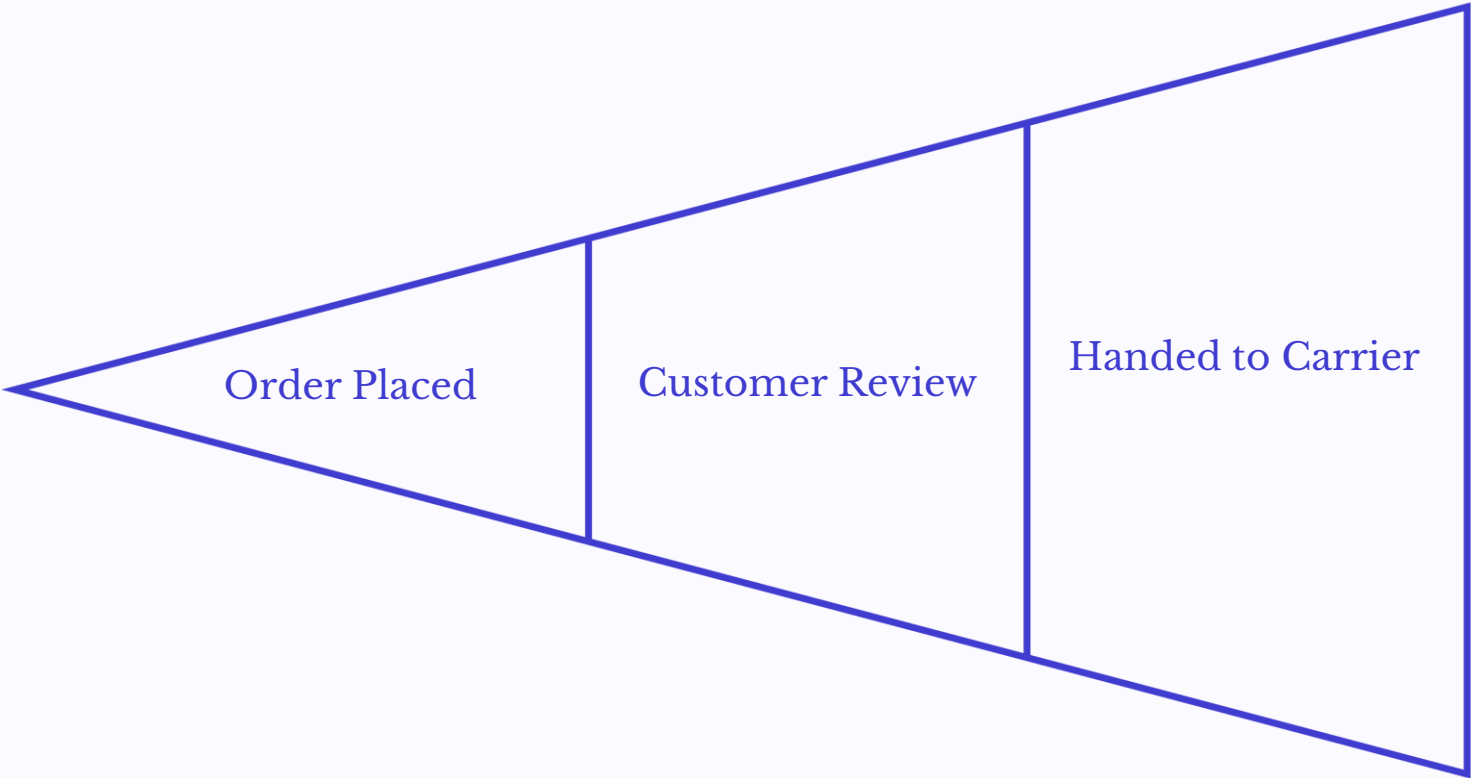
The Business Question

Where do customers drop off between placing an order and receiving it?

Funnel Analysis tracks how orders move through each stage — and quantifies where, why, and at what cost they fall out.

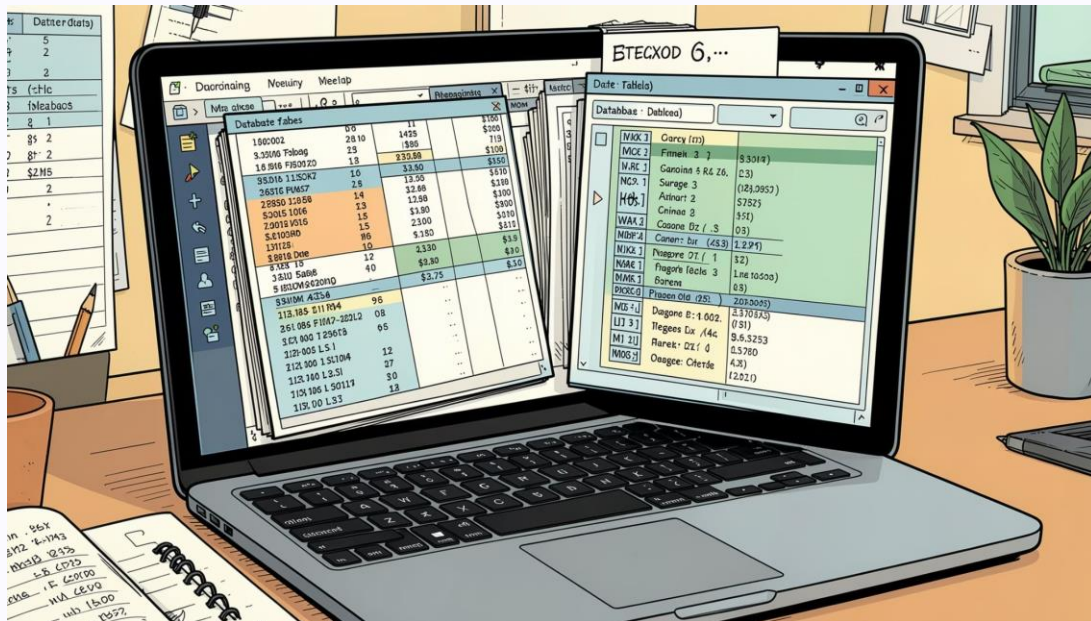


The 5-Stage Order Funnel



At every transition, some orders drop off. This project maps each stage, measures conversion, and identifies the biggest friction points.

Data Sources



Orders

Timestamps for each funnel stage

Customers

City, state, and location data

Items & Products

Price, category, and SKU details

Payments & Reviews

Method, amount, and 1–5 scores

Tools & Workflow



MS SQL Server

Schema design, data loading, and business analysis queries



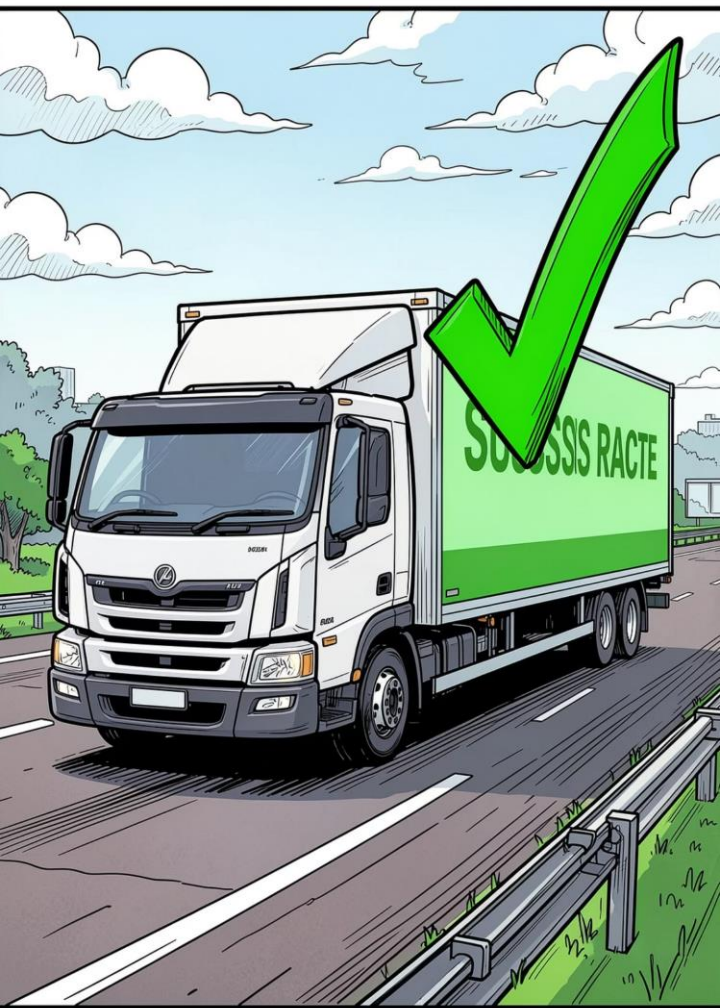
Python

Exploratory Data Analysis in Google Colab / Jupyter



Power BI

Interactive reporting dashboard for stakeholders



97% Delivery Success Rate

97%

Orders Delivered

Near-perfect fulfillment rate

74%

Credit Card Usage

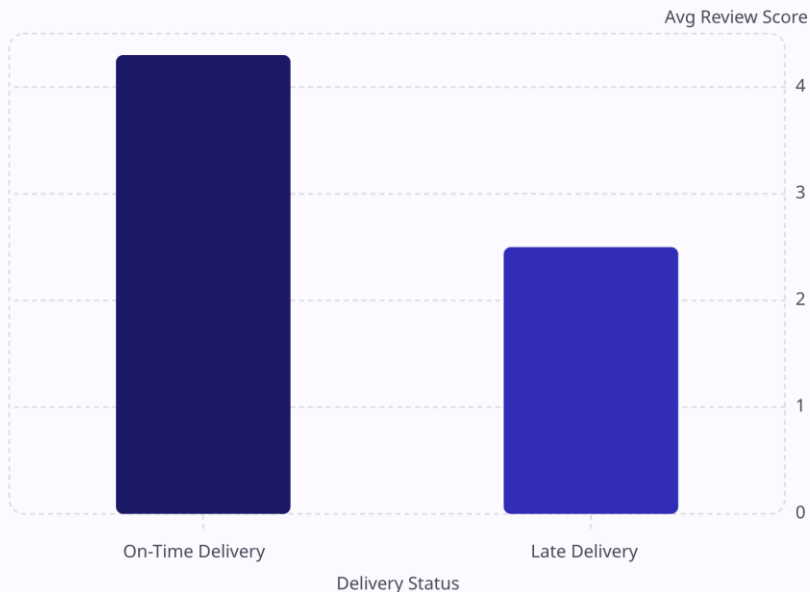
Highest delivery rate by payment
type

90%+

One-Time Buyers

Very low repeat purchase rate

Late Deliveries Crush Review Scores

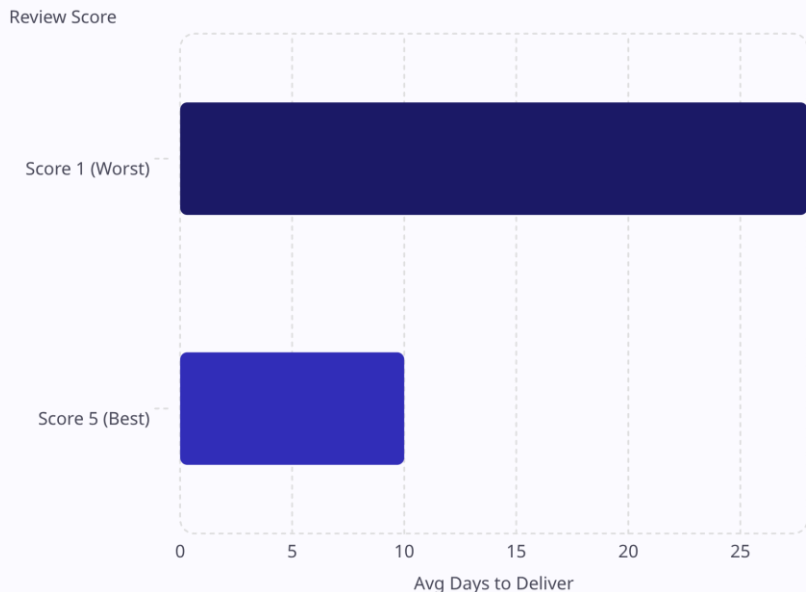


The Gap Is Staggering

On-time orders average **4.3 stars**. Late orders average just **2.5 stars** — a **1.8-point drop** driven entirely by delivery speed.

- ❏ Delivery performance is a direct lever for customer satisfaction.

Review Score vs. Delivery Time

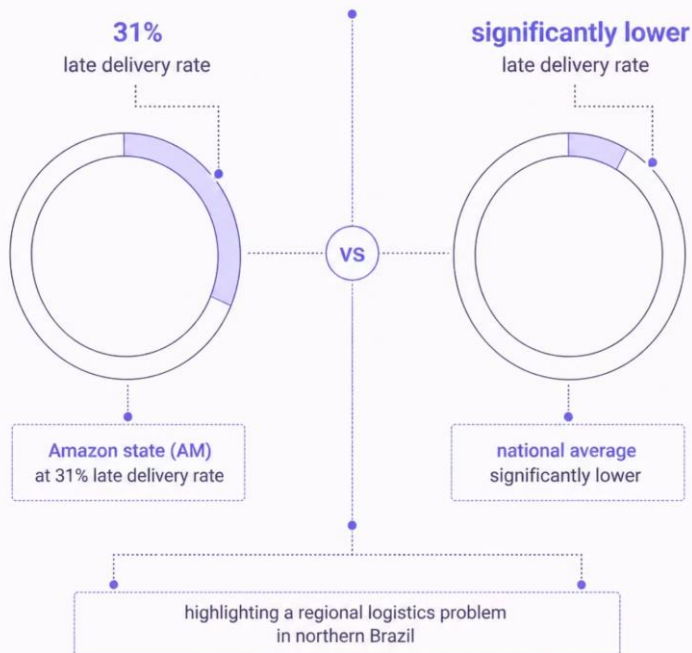


Speed = Satisfaction

1-star orders took **28 days** on average to deliver — nearly **3× longer** than 5-star orders at just **10 days**.

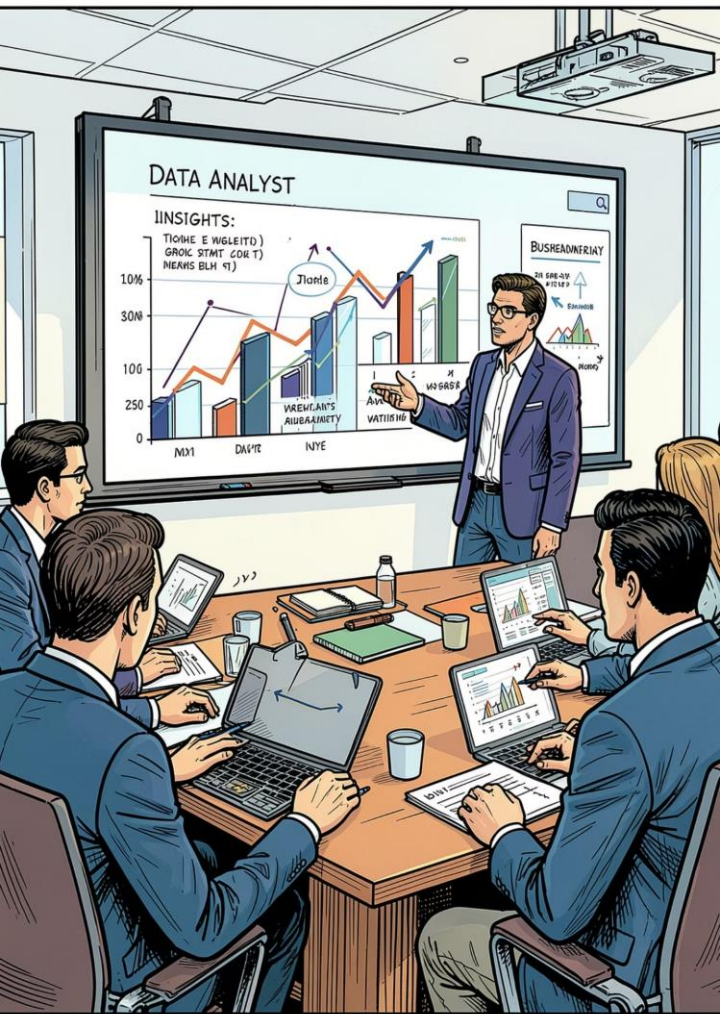
Delivery speed is the strongest predictor of review score.

Regional Logistics Gap: Amazon State (AM)



Amazon State Stands Out

State of Amazonas (AM) recorded a **31% late delivery rate** — far above the national average. Geographic isolation and infrastructure challenges make this a clear logistics problem requiring targeted intervention.



Key Takeaways

🚚 Fix Late Deliveries

Biggest driver of low review scores

📍 Target Amazon State

31% late rate — regional logistics priority

💳 Promote Credit Card

74% usage, highest delivery success

🔄 Boost Retention

90%+ one-time buyers — loyalty gap